EE3621: Digital Signal Processing | III B.Tech EEE

Faculty Reference Document — Textbook Replacement

PREFACE FOR FACULTY

The Frequency-Sampling Method designs an FIR filter by directly sampling the desired continuous frequency response $H_d(e^{j\omega})$ at N uniform frequency points $\omega_k = \frac{2\pi k}{N}$ and computing the filter coefficients $h[n]$ via Inverse DFT.

Pedagogical Strategy:

1. Formulate the Frequency-Sampling equations: $H[k] = |H_d(e^{j2\pi k/N})|e^{-j\frac{N-1}{2}\frac{2\pi k}{N}}$.
2. Apply the IDFT synthesis formula to calculate real impulse response coefficients $h[n]$.
3. Explain the primary drawback of basic frequency sampling: Between sampled points, the frequency response ripples significantly, yielding poor stopband attenuation ($\approx -16 \text{ dB}$).
4. Demonstrate **Transition Band Optimization**: By placing one or two unconstrained optimization variables $T_1, T_2 \in (0, 1)$ in the transition band, stopband attenuation is dramatically increased to over -45 dB (1 transition sample) or -75 dB (2 transition samples).

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Sample** desired frequency responses $H_d(e^{j\omega})$ with proper linear-phase phase assignments.
2. **Compute** FIR filter coefficients $h[n]$ via IDFT formulation.
3. **Optimize** transition band samples to maximize stopband attenuation.
4. **Compare** the Frequency-Sampling method with the Windowing method.

2. MATHEMATICAL FOUNDATIONS

2.1 Frequency-Sampling Design Equations

Let $H[k] = H_d(e^{j\frac{2\pi k}{N}})$. For a linear-phase FIR filter of length N:

$$H[k] = |H[k]|e^{j\angle H[k]}, \quad \angle H[k] = -\left(\frac{N-1}{2}\right)\frac{2\pi k}{N}$$

The impulse response is computed by the IDFT:

$$h[n] = \frac{1}{N}\sum_{k=0}^{N-1} H[k]e^{j\frac{2\pi kn}{N}}$$

Exploiting Hermitian symmetry ($H[N-k] = H^*[k]$):

- **For N Odd:**

$$h[n] = \frac{H[0]}{N} + \frac{2}{N}\sum_{k=1}^{(N-1)/2} |H[k]| \cos\left[\frac{2\pi k}{N}\left(n - \frac{N-1}{2}\right)\right]$$

- **For N Even:**

$$h[n] = \frac{H[0]}{N} + \frac{2}{N}\sum_{k=1}^{N/2-1} |H[k]| \cos\left[\frac{2\pi k}{N}\left(n - \frac{N-1}{2}\right)\right] + \frac{H[N/2]}{N}(-1)^n$$

2.2 Transition Sample Optimization Values

Number of Transition Samples	Optimized Sample Values	Resulting Stopband Attenuation A_s
0 (No transition band)	None ($H[k] \in 0, 1$)	-16 dB
1 Transition Sample	$T_1 \approx 0.38$	-45 dB
2 Transition Samples	$T_1 \approx 0.59, ; T_2 \approx 0.11$	-75 dB
3 Transition Samples	$T_1 \approx 0.70, ; T_2 \approx 0.25, ; T_3 \approx 0.02$	-95 dB

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3. WORKED NUMERICAL EXAMPLES

Example 24.1: Frequency-Sampling Lowpass Filter Design

Problem: Design a 7-point ($N = 7$) linear-phase FIR lowpass filter with frequency samples:

$$|H[k]| = 1, 1, 0, 0, 0, 0, 1 \quad \text{for } k = 0, 1, 2, 3, 4, 5, 6$$

Solution:

- Length $N = 7$ (Odd), $\tau = \frac{7-1}{2} = 3$.
- Non-zero samples: $|H[0]| = 1, ; |H[1]| = |H[6]| = 1$.
- Using the real summation formula:

$$h[n] = \frac{H[0]}{7} + \frac{2}{7}|H[1]| \cos\left[\frac{2\pi(1)}{7}(n-3)\right] = \frac{1}{7} + \frac{2}{7}\cos\left[\frac{2\pi}{7}(n-3)\right]$$

  - $h[3] = \frac{1}{7} + \frac{2}{7}\cos(0) = \frac{3}{7} \approx 0.4286$
  - $h[2] = h[4] = \frac{1}{7} + \frac{2}{7}\cos\left(\frac{2\pi}{7}\right) = \frac{1+2(0.6235)}{7} = \frac{2.2470}{7} \approx 0.3210$
  - $h[1] = h[5] = \frac{1}{7} + \frac{2}{7}\cos\left(\frac{4\pi}{7}\right) = \frac{1+2(-0.2225)}{7} = \frac{0.5550}{7} \approx 0.0793$
  - $h[0] = h[6] = \frac{1}{7} + \frac{2}{7}\cos\left(\frac{6\pi}{7}\right) = \frac{1+2(-0.9010)}{7} = \frac{-0.8019}{7} \approx -0.1146$

$$h[n] = -0.\underset{\uparrow}{1}146, 0.0793, 0.3210, 0.4286, 0.3210, 0.0793, -0.1146$$

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4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) Derive the Frequency-Sampling design formula for an odd-length linear-phase FIR filter. *(8 Marks)* **(b)** Explain why transition band samples are introduced in the Frequency-Sampling method. How do they affect the passband ripple and stopband attenuation? *(7 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
  - Mathematical derivation using IDFT and Euler's formula *(8 Marks)*
- **Part (b):**
  - Explanation of interpolation ripple between sharp 1-to-0 transitions *(3 Marks)*
  - Role of transition samples T_1, T_2 in smoothing frequency gradient and boosting A_s from 16 dB to $> 45 \text{ dB}$ *(4 Marks)*

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5. PYTHON VERIFICATION SCRIPT

```
import numpy as np

N = 7
H = np.array([1, 1, 0, 0, 0, 0, 1], dtype=complex)
# Apply linear phase
k = np.arange(N)
H_phase = H * np.exp(-1j * ((N - 1) / 2) * 2 * np.pi * k / N)
h = np.fft.ifft(H_phase).real
print("Frequency Sampling Coefficients h[n]:", np.round(h, 4))
```